

Architecture Design Document: Project Oracle

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System Status: Design Phase (Prototype v0.1)

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1. Introduction & Objectives

The Architecture Design Document (ADD) provides a high-level technical map of the Project Oracle ecosystem. The system is designed to provide low-latency AI inference (Vision & Language) over streaming financial data, integrated into the TradingView Charting Library.

2. System Architecture (C4 Model – Level 2)

```
[CLIENT LAYER] |-- TradingView Library (JS) |-- Dante Oracle Side-Panel (React) |--  
WebSocket Client (Signal Ingest) [GATEWAY LAYER (Go/gRPC)] |-- Load Balancer (Nginx/Envoy)  
|-- Authentication (JWT/Auth0) |-- Signal Router [INTELLIGENCE LAYER (Python/Triton)] |--  
Dwell-Time Optimization Engine |-- LLM (PineCoder-70B) → [Quantized 4-bit] |-- Vision CNN  
(ChartTopology-V1) [DATA LAYER] |-- Kafka (Message Broker) |-- ClickHouse (Time-Series /  
Backtest) |-- Redis (Cache Signal Hot-State)
```

3. Component Specifications

3.1 Pipeline Ingestion (The "Inflow")

Uses [Go](#) to handle 100k+ concurrent WebSockets. Ingests raw OHLCV from Binance/CME, normalizes the data, and pushes it to Kafka topics for consumption by the Intelligence Layer.

3.2 Vision Analysis Engine

A **PyTorch-based CNN** trained on chart screenshots (1024×1024). It performs "Pixel-to-Pattern" mapping, bypassing historical math lag. Output is a JSON object containing pattern coordinates and confidence scores.

4. Data Schema & Messaging

4.1 Internal Signal Message (Protobuf)

```
message MarketSignal { string symbol = 1; int64 timestamp = 2; SignalType type = 3; //
VISION, LLM_PINE, SENTIMENT float confidence = 4; map<string, string> metadata = 5; // e.g.
"script_v5_body" }
```

5. Scaling & Infrastructure

Infrastructure	Solution	Justification
Cloud Provider	AWS / GCP	Required for H100/L40S GPU availability.
Orchestration	Kubernetes (EKS)	Horizontal scaling based on GPU utilization.
Storage	S3 + ClickHouse	Storage of massive historical pattern datasets.

6. Security Architecture

- **Inference Isolation:** AI models run in sandboxed containers to prevent prompt injection from affecting the core data pipeline.
- **End-to-End Encryption:** TLS 1.3 for all streaming data; strategies encrypted with user-specific keys.
- **Audit Logging:** Every AI-generated Pine Script is logged and hashed for compliance auditing.

7. Technical Risk Assessment

- Model Drift

Risk Component	Mitigation
GPU Inference Latency	NVIDIA TensorRT optimization + caching frequent queries.
UDF API Throughput	Custom Binary adapter for TradingView datafeed.
Daily calibration loops against live market results.	